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Structural insights and predictive screening of ion transport in Li-rich alloys via neuroevolution potentials

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Fast and efficient lithium-ion transport is crucial for high-performance rechargeable batteries but remains poorly understood in complex alloy anodes due to the coexistence of multiple competing diffusion mechanisms. A key challenge is disentangling how defect type, local coordination, and structural connectivity collectively govern migration barriers, which has so far limited the development of generalizable design rules. Here, we develop neuroevolution potentials (NEPs), a machine-learning framework trained to near-density functional theory (DFT) accuracy, providing both reliability and transferability for large-scale simulations of Li diffusion in Li-In and Li-Sn alloys. Using NEPs, we identify three primary factors controlling Li mobility: the dominant defect carrier (Li vacancy, Li interstitial, or metal vacancy), the connectivity of low-barrier diffusion pathways, and the local atomic environment, with migration distance and charge redistribution as the most critical determinants of diffusion barriers. Li diffusion shows strong structural inheritance across intermetallics, where the crystal backbone dictates long-range connectivity and chemical composition fine-tunes local behavior. The favorable diffusion structural frameworks revealed by NEPs can be extended to other Li-rich intermetallics, enabling predictive screening of fast-diffusing phases. This work establishes a predictive framework for designing next-generation alloy anodes with high ionic conductivity.

Rechargeable lithium-ion batteries (LIBs) are indispensable to portable electronics, electric vehicles, and grid-scale energy storage, forming a cornerstone of the global transition toward sustainable energy^{1,2}. Achieving higher energy density, faster charging, and longer cycle life critically depends on the mobility of lithium ions within electrode materials. Beyond thermodynamics and interfacial chemistry, lithium diffusion represents a fundamental rate-limiting factor governing power performance and overall device efficiency^{3,4}. Elucidating the atomistic mechanisms of Li migration is therefore essential for optimizing electrode design and extending the frontiers of energy storage technologies^{5,6}.

Extensive studies have characterized lithium diffusion across various electrode classes, including from intercalation-type oxides, alloying-based materials, and conversion-based systems^{7–10}. These investigations collectively reveal that Li kinetics are highly sensitive to the local chemical and structural environment, which depends on factors, such as composition¹¹, phase¹², and lithiation state¹³. Consequently, developing a systematic methodology that links local chemical and structural environments to diffusion pathways and migration barriers across different material classes is

valuable for identifying the key factors governing Li transport and guiding electrode design. For intercalation-type oxide electrodes and oxide-type solid-state electrolytes, previous studies have established that parameters, such as the valence of the transition-metal cation, the inter-cation distances, local coordination geometry, and polyhedral connectivity can affect lithium-ion mobility^{14–16}. In contrast, analogous studies in alloy-based systems remain relatively limited, leaving a gap in transferable insights across material classes. Advancing understanding in this area would enable more predictive tuning of ion transport behavior and support the rational design of high-performance alloy electrodes.

Atomistic simulations provide indispensable insight into lithium transport at the microscopic level^{17,18}. The nudged elastic band (NEB) method can determine migration barriers along predefined paths, while heuristic approaches, such as bond-valence and Ewald-energy analyses can map plausible routes but cannot quantify conductivity^{19,20}. Molecular dynamics (MD) simulations capture finite-temperature diffusion and mechanistic diversity, yet *ab initio* MD (AIMD) is computationally prohibitive for exploring alloy phase diagrams, metastabilities, and long

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timescales²¹. Recent advances in machine-learning potentials (MLPs) offer a promising route forward, delivering near-density functional theory (DFT) accuracy at orders-of-magnitude lower cost^{22,23}. MLPs have already been applied to Li diffusion^{24–26}, phase stability^{27–29}, nucleation³⁰, and lithiation kinetics^{31–33}, and, in combination with data-driven models, they present new opportunities for constructing transferable structure-performance descriptors^{34,35}. Nevertheless, a unified, quantitatively validated framework that connects carrier type (Li vacancies, Li interstitials, or metal vacancies), local coordination/electronic responses, and migration barriers across complex alloy phases remains to be established^{36,37}.

Alloy anodes represent a compelling strategy to overcome the limitations of lithium metal, allowing Li ions to diffuse into host structures and form stable alloy phases that mitigate dendrite formation and interfacial degradation³⁸. Among them, Li-In and Li-Sn alloys serve as ideal model systems due to their rich phase diagrams encompassing multiple stable and low-energy metastable compounds, diverse structural configurations, and pronounced phase transformations during (de)lithiation^{39,40}. These characteristics enable systematic interrogation of how atomic-scale environments influence Li kinetics. Their practical relevance is reinforced by favorable interfacial stability in solid-state batteries and high-rate performance in specific phases. Spanning broad structural diversity within two chemically related binaries, Li-In and Li-Sn provide the necessary scope to extract transferable mechanistic rules while maintaining interpretability.

In this work, we develop neuroevolution potentials (NEPs) for Li-In and Li-Sn alloys through a workflow that integrates large-scale structural searches and molecular-dynamics-driven data generation. Using this framework, we (i) map phase stability and uncover previously unreported low-energy structures, and (ii) elucidate the atomic-scale mechanisms of Li diffusion, identifying vacancy- and interstitial-mediated pathways and quantifying how network connectivity governs long-range mobility. Our analysis reveals that the topology of low-energy diffusion channels, rather than individual migration barriers, dominates Li transport, and that crystal framework and local coordination determine ionic mobility, with chemical composition serving a secondary fine-tuning role. Favorable diffusion pathways exhibit strong structural inheritance across Li-rich intermetallics, providing a mechanistic basis for rational design of fast-diffusing alloy anodes. Collectively, this work establishes a unified grounded framework for understanding and predicting Li transport in complex alloys, supporting the design of next-generation electrodes with high ionic conductivity.

Results

Machine-Learning Potential Development

MLPs for Li-In and Li-Sn alloys are developed using the NEP framework (parameters in Table S1) implemented in GPUMD^{41,42}. In this framework, the total energy of the system, U , is expressed as a sum of atomic site energies U_i , which depend on the local environment of each atom within a finite cutoff radius. By construction, this approach allows efficient scaling to larger systems while preserving DFT-level accuracy, where the reliability of predictions depends critically on the comprehensiveness and representativeness of the training dataset. The initial training datasets are generated by combining two complementary strategies (Fig. 1a). First, equilibrium AIMD simulations are performed at temperatures from 300 to 1800 K on known structures obtained from the Materials Project and OQMD databases^{43,44}, with snapshots sampled at regular time intervals to capture thermally accessible atomic configurations. Second, isotropic structural perturbations are introduced by scaling lattice parameters by factors of 0.8–1.2, thereby sampling non-equilibrium configurations to enhance coverage of the potential energy landscape. Principal component analysis (PCA)⁴⁵ is used to evaluate the diversity and structural distribution of the constructed dataset. As shown in Fig. 1a, projection onto a two-dimensional PCA space reveals that local atomic environments for each composition are well represented; however, the sampled configurations occupy a relatively narrow region of the potential energy landscape. This indicates that, although the initial dataset captures key structural variations, additional sampling is required to achieve broader and more robust coverage of the configurational space.

To further enhance structural diversity, especially for configurations relevant to Li diffusion and phase transformations, the NEP model is integrated with CALYPSO for random structure prediction in an iterative workflow⁴⁶. This integration leverages the excellent balance of NEP between DFT-level accuracy and computational efficiency^{23,47,48}. A diverse structural dataset is generated using CALYPSO, whose efficient global search capabilities enhance the coverage and robustness of the NEP training process^{28,49}. The initial dataset shown in the left panel of Fig. 1a is used as the training set to construct the NEP model, which is subsequently employed for structural relaxations in the first generation of CALYPSO exploration. In each cycle, 2500 candidate structures are generated, and the most distinct ones are selected using farthest-point sampling, a strategy that maximizes geometric diversity while minimizing redundancy⁴². These selected structures are subsequently evaluated by DFT calculation to obtain energies, forces, and virials, which are added to the training set before retraining the NEP model (the right panel of Fig. 1a). This process is repeated until the model accurately predicted all candidate structures, resulting in final datasets of 2612 Li-In and 3106 Li-Sn structures. PCA analysis confirms a significant increase in configurational coverage compared with the initial datasets.

Validation of the NEP models is performed against DFT calculations. To evaluate the performance of the final NEP model, we first check its ability to reproduce DFT results. Figure 1b and S1 shows good agreement between NEP and DFT results for energies, forces, and virials across a wide range of compositions and structures. For Li-In (Li-Sn) alloys, the root-mean-square errors (RMSEs) for energies, forces, and virial tensors in the training set are 11 (16) meV/atom, 64 (67) meV/Å, and 45 (71) meV/atom, respectively. For an independent test set of 2000 randomly generated Li-In (Li-Sn) structures, the RMSE are 7 (8) meV/atom, 46 (64) meV/Å, and 42 (46) meV/atom. Furthermore, NEP-MD simulations reproduce AIMD atomic trajectories, radial distribution functions (RDF), phonon spectra, energy-volume relations, and lattice parameters (deviations $\leq 0.9\%$), confirming that NEP reliably captures both local bonding and long-range structural correlations (Fig. S2 and Table S2). Collectively, these results demonstrate that the developed NEP models are accurate, transferable, and suitable for large-scale MD simulations aimed at systematically exploring Li diffusion pathways, migration mechanisms, and kinetics in complex Li-In and Li-Sn alloys.

Phase Stability and Structural Evolution of Li-In and Li-Sn Alloys

The phase diagrams of the Li-In and Li-Sn alloy systems are revisited through large-scale structure searches using both CALYPSO and USPEX⁵⁰, with candidate energetics efficiently screened by the NEP model. The formation energy of each compound is defined as the energy difference between the alloy and the corresponding amounts of elemental solid Li and M (M = In or Sn), normalized per atom. Figure 2a, b present a comparison of formation energies for low-energy structures in the Li-In and Li-Sn systems as predicted by NEP and DFT, with RMSEs of 0.017 and 0.023 eV/atom, respectively. To account for these small deviations, a threshold of 0.050 eV/atom above the convex hull is adopted. All NEP-predicted structures within this range are re-optimized using DFT, following established practices in MLP-based structure prediction^{28,51}. For high-symmetry structures within 0.010 eV/atom of the hull, zero-point energy (ZPE) corrections are additionally incorporated to their important influence on structural stability of competing low-energy phases containing light elements. Structures lying further from the convex hull are excluded from refinement, as their energies are sufficiently high to rule them out as ground-state candidates. In total, 1172 Li-In and 1077 Li-Sn structures are examined. As shown in Fig. 2c and S3a, the NEP model reliably reproduces the DFT convex-hull landscapes, with only minor deviations in equilibrium phases and formation energies.

A key outcome of our structure searches is the prediction of several previously unreported stable and low-energy metastable phases in the Li-In and Li-Sn systems, which extend the known compositional and structural landscape of these alloys. As shown in Fig. 2e and Figure S3b–g, our

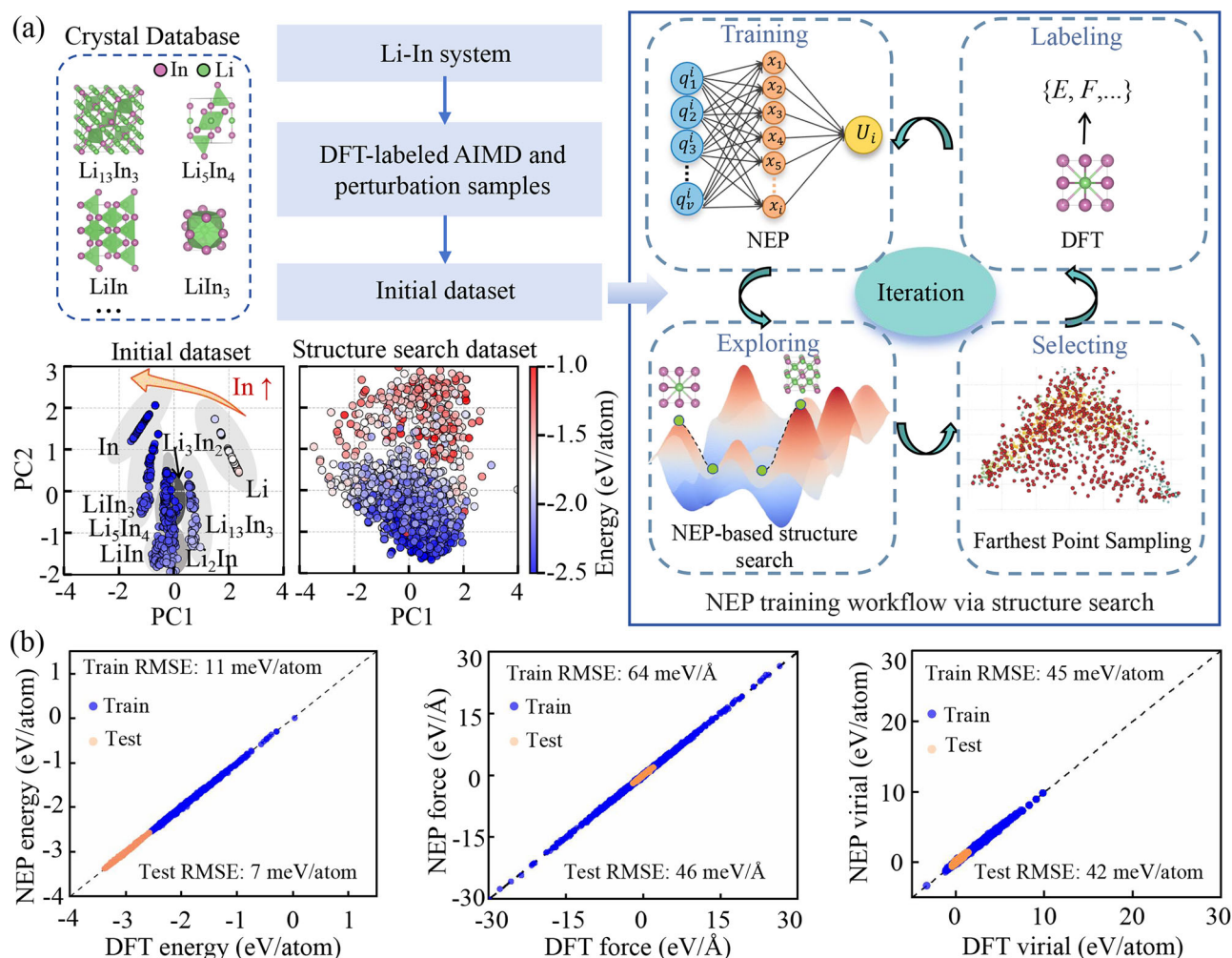


Fig. 1 | Development and validation of the neuroevolution potentials (NEP) model for Li-In systems. **a** Workflow for developing the NEP model for Li-In systems, along with the principal component analysis (PCA) of the initial and structure-search-generated datasets. PC1 and PC2 represent the first and second

principal components. The reduced dimensions do not carry explicit physical meaning but represent the relative distribution and clustering of configurations in the low-dimensional space. **b** Comparison of NEP-predicted energy, force, and virial tensor values with density functional theory (DFT) results for Li-In systems.

calculations capture the majority of experimentally reported equilibrium phases^{52,53}, validating the reliability of the NEP model. Two unreported stable compounds are disclosed in the Li-In system, Li₇In₄ (*C2/m*), and Li₁₇In₁₅ (*Pmm2*). In addition, several low-lying metastable structures with energies within 10 meV/atom above the convex hull are identified (Figs. S4 and S5), including Li₁₃In₅ (*Pm*, 6 meV/atom), Li₁₁In₄ (*P3m1*, 2 meV/atom), and Li₁₉In₅ (*Pmmm*, 7 meV/atom), as well as Li₇Sn₆ (*C2/m*, 7 meV/atom) and Li₃Sn₂ (*Cmcm*, 9 meV/atom). All of these phases are dynamically stable (Fig. S6), suggesting their potential synthesizability. Collectively, these predicted stable and metastable phases expand the known phase diagrams of Li-In and Li-Sn alloys and provide guidance for future experimental exploration.

The compositional evolution of the M (In/Sn) sublattice provides mechanistic insight into bonding characteristics across the Li-In and Li-Sn alloy systems. With increasing Li concentration, the In sublattice evolves from three-dimensional (3-D) frameworks and two-dimensional (2-D) layers to one-dimensional chains, disconnected layers, or dimers, eventually forms isolated atoms. In compounds, such as Li₃In₂, Li₅In₄, and LiIn, the In sublattice adopts six-membered ring networks. According to the Zintl-Klemm concept and the 8-N rule, when the Li:In ratio approaches 1:1, In exhibits bonding behavior similar to group-IV elements. Likewise, when the Li:Sn ratio is near 1:1, Sn adopts bonding characteristics analogous to Sb,

forming four-membered ring frameworks. Bader charge, electron localization, and density-of-states analyses indicate that Li consistently donates electrons to In or Sn in both alloy systems (Figs. S7–S8). Although the degree of charge transfer slightly decreases under Li-rich conditions, the interaction remains predominantly ionic.

The volume evolution of In and Sn alloys during lithiation is shown in Fig. S9. Upon full lithiation, Sn and In expand by ~250% (Li₂₂Sn₅) and ~260% (Li₁₃In₃), respectively, consistent with previous theoretical and in situ TEM results^{54,55}. Predicted expansions at intermediate lithiation stage also fall within this range, indicating that the calculated volume changes are broadly consistent with experimental observations.

The predicted electrochemical properties reflect these structural configurations and the relative stability of the alloys. Using the convex hull data, the average voltages of the Li-In and Li-Sn systems are evaluated, including the two previously unknown stable phases (Fig. 2d and S10). The resulting voltage plateaus closely match those of experimentally reported compounds⁵⁶. For instance, Li₁₇In₁₅ exhibits a voltage plateau of 0.47 V, comparable to LiIn (0.48 V) and Li₅In₄ (0.46 V), suggesting it may form during electrochemical cycling. The Li-In system generally exhibits lower average voltages than Li-Sn, mainly due to the more negative formation energies of Li-Sn phases indicate greater energy release during alloying, leading to higher thermodynamic stability and higher average voltage.

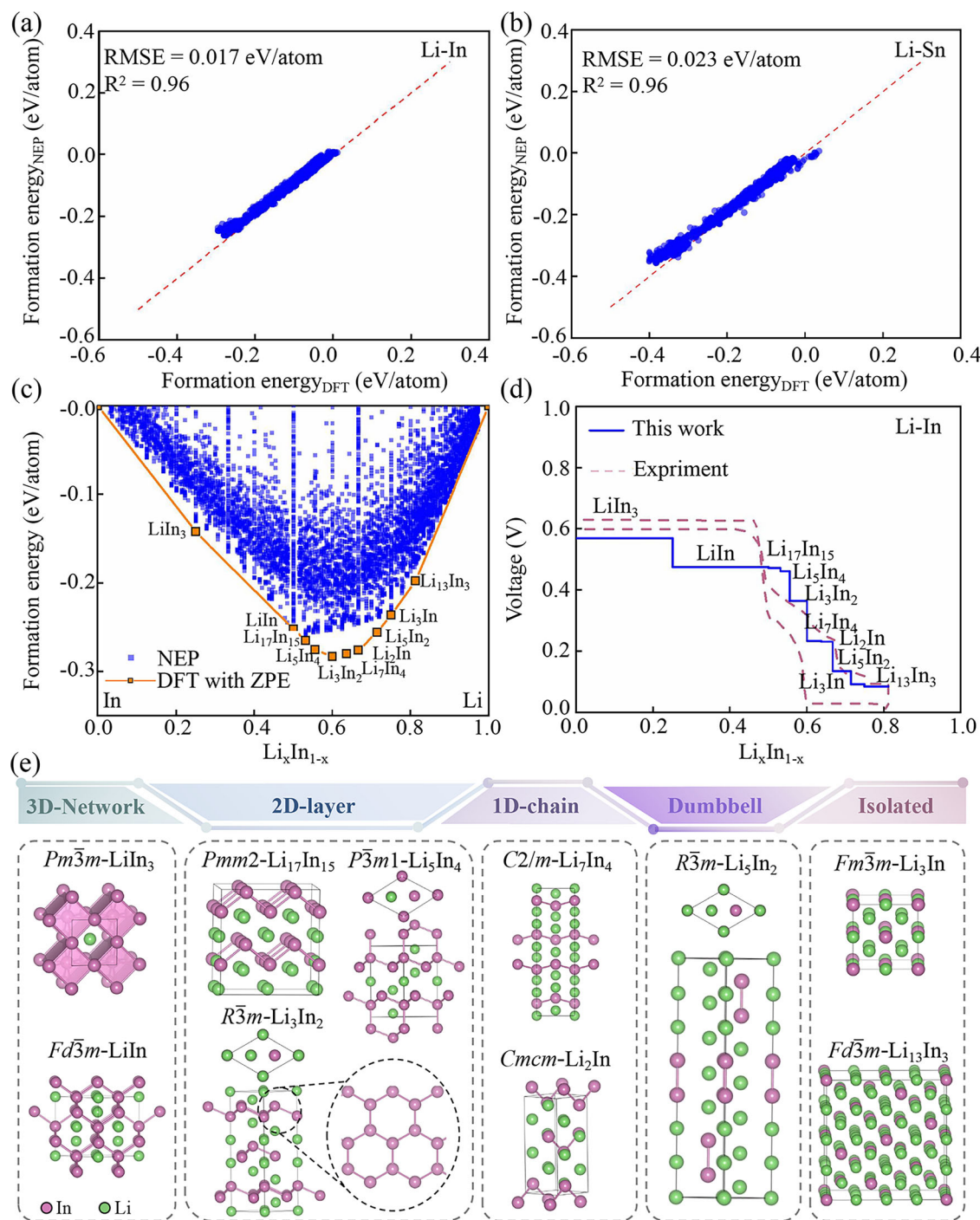


Fig. 2 | Thermodynamic stability and electrochemical properties of Li-In compounds. **a, b** Comparison of formation energies between NEP and DFT for low-lying structures uncovered by structure searches in Li-In and Li-Sn systems. **c** Formation energies of Li-In compounds relative to elemental solid Li and In. **d** Calculated

average voltages versus lithium for the stable Li-In structures, compared with experimental data⁵⁶. Here, x denotes the Li content in $\text{Li}_x\text{In}_{1-x}$. **e** Thermodynamically stable Li-In compounds with Li and In atoms in green and mauve, respectively.

Atomic-Scale Mechanisms and Diffusion Pathways of Li in Li-In and Li-Sn Alloys

Understanding the atomic-scale mechanisms governing Li-ion transport in Li-In and Li-Sn alloys is crucial for elucidating their electrochemical behavior and optimizing battery performance. Li diffusion in these systems primarily occurs via interstitial and vacancy-mediated migration, each associated with distinct dominant carriers. To clarify their contributions, we performed NEP-MD simulations under three representative conditions—M (In/Sn) vacancies, Li vacancies, and Li interstitials—using a uniform 1% defect concentration. To validate the reliability of the NEP, we

compared the Li self-diffusion coefficients and activation energies obtained from pulsed field gradient nuclear magnetic resonance spectroscopy (PFG-NMR) measurements⁵⁷ of LiIn with those predicted by the NEP (Fig. 3a and S11). The calculated Li-vacancy diffusion coefficient ($6.0 \pm 0.2 \times 10^{-7} \text{ cm}^2/\text{s}$) and activation energy ($0.05 \pm 0.002 \text{ eV}$) at 300 K are in good agreement with the experimental values ($5.5 \pm 1.0 \times 10^{-7} \text{ cm}^2/\text{s}$ and $0.068 \pm 0.037 \text{ eV}$). Besides, incorporating the uncertainty does not alter the mechanistic interpretation: Li vacancies consistently remain the dominant carriers of LiIn, and the relative ranking of diffusion performance among the materials is preserved.

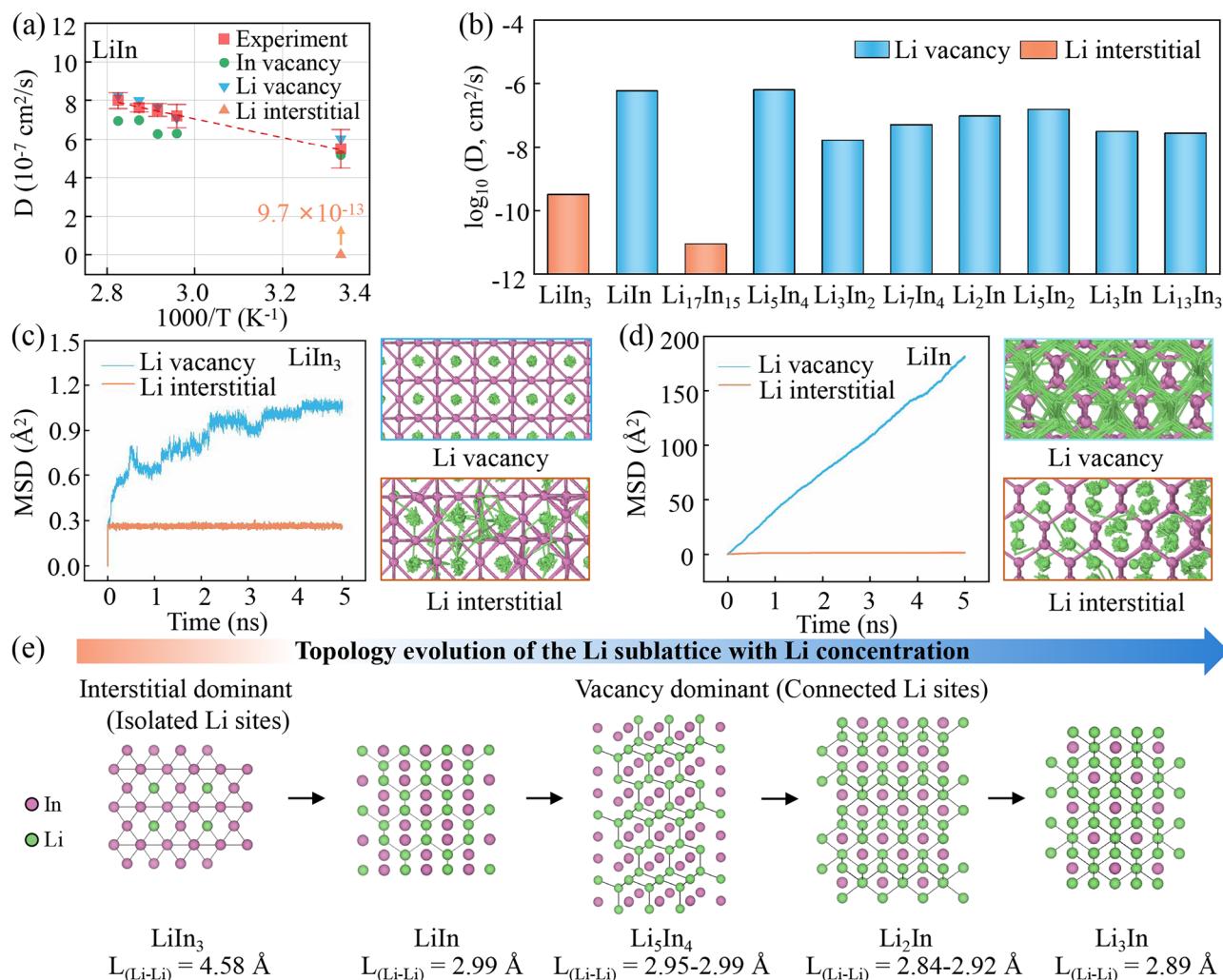


Fig. 3 | Lithium-ion transport mechanisms in Li-In alloys. **a** Self-diffusivity (D) of Li^+ in LiIn ($Fm\bar{3}m$) with 1% Li defects as a function of temperature, obtained from NEP-MD simulations and compared with experimental self-diffusion data measured using pulsed field gradient nuclear magnetic resonance spectroscopy (PFG-NMR)⁵⁷. **b** Li diffusion coefficients and dominant carriers in stable Li-In structures at

300 K. **c, d** Mean Square Displacement (MSD) profiles and Li-atom trajectories (green line) at 300 K are shown for LiIn_3 and LiIn with 1% Li vacancy and 1% Li interstitial atoms. **e** Mechanistic transition of Li diffusion in stable Li-In alloys from vacancy- to interstitial-dominated transition with increasing Li concentration, correlated with Li sublattice network connectivity.

For stable Li-In structures, Fig. S12 identifies the dominant charge carriers by comparing the MSD values at 300 K and presents the temperature-dependent diffusion coefficients and activation energies, with the maximum diffusion coefficients and corresponding dominant carriers at 300 K shown in Fig. 3b. For LiIn_3 and $\text{Li}_7\text{In}_{15}$, since the MSD could not be fitted linearly at 300 K, we estimated their diffusion coefficients using high-temperature extrapolation⁵⁸. In LiIn_3 and LiIn , different carriers govern Li diffusion, and their nature strongly influences the diffusion behavior (Fig. 3c, d). This carrier dependence is directly linked to the connectivity of the Li sublattice network (Fig. 3e). For Li_xIn with $x < 1$, interstitial Li dominates due to isolated Li atoms with In nearest neighbors (3.24 Å) and Li-Li distances of 4.58 Å; for $x > 1$, Li vacancies dominate as Li atom form a continuous 3D network. $\text{Li}_7\text{In}_{15}$ is an exception, where interstitial Li remains dominant despite $x > 1$ (Fig. S13). These transitions agree with Galvanostatic Intermittent Titration Technique experiments and electrochemical impedance spectroscopy⁵⁹. We also note that, for Li_3In_2 , experimental studies support vacancy-mediated diffusion⁵⁹, whereas previous NEB calculations reported slightly higher barriers for vacancy diffusion than for interstitial diffusion⁶⁰. Our NEP-MD results, in agreement with experimental observations, demonstrate that vacancy-mediated Li species can indeed undergo long-range diffusion (Fig. S14). This agreement may arise from the ability of NEP-MD to sample spontaneous diffusion events

and to capture extended migration trajectories beyond the constraints of predefined NEB pathways. Among the studied phases, LiIn ($Fm\bar{3}m$) and Li_5In_4 exhibit diffusion coefficients on the order of $10^{-7} \text{ cm}^2/\text{s}$ at room temperature, demonstrating excellent ionic mobility. No diffusion dominated by In vacancies is observed in any Li-In configuration. Under Li-vacancy-mediated conditions, In atoms primarily vibrate around their equilibrium positions due to the high migration barriers and do not occupy Li sublattices, thus posing no hindrance to Li diffusion. Sn atoms in the Li-Sn system exhibit analogous behavior (Figs. S15–S17). In structures dominated by interstitial Li, In and Sn atoms can exhibit cooperative motion with Li (such as LiIn_3), contributing to overall diffusion.

Local lattice rearrangements and diffusion pathway reconstruction underlie these transitions. To elucidate the microscopic origin, we combined NEB calculations with NEP-MD trajectory analysis to quantify diffusion characteristics at the atomic scale. At low Li content (LiIn_3), vacancy-mediated diffusion follows LiIn_{10} tetrakaidecahedron $\rightarrow$ LiIn_4 planar quadrilateral $\rightarrow$ LiIn_{10} tetrakaidecahedron (Path I, Fig. 4a and S18), with an energy barrier (ΔE) of 2.27 eV. In contrast, interstitial Li migrates via a knock-off mechanism, cooperatively moving with neighboring In atoms to hop from LiIn_4 to LiIn_7 polyhedra, with $\Delta E = 0.005 \text{ eV}$, making interstitial Li the most efficient diffusion carrier in this configuration (Fig. 4c).

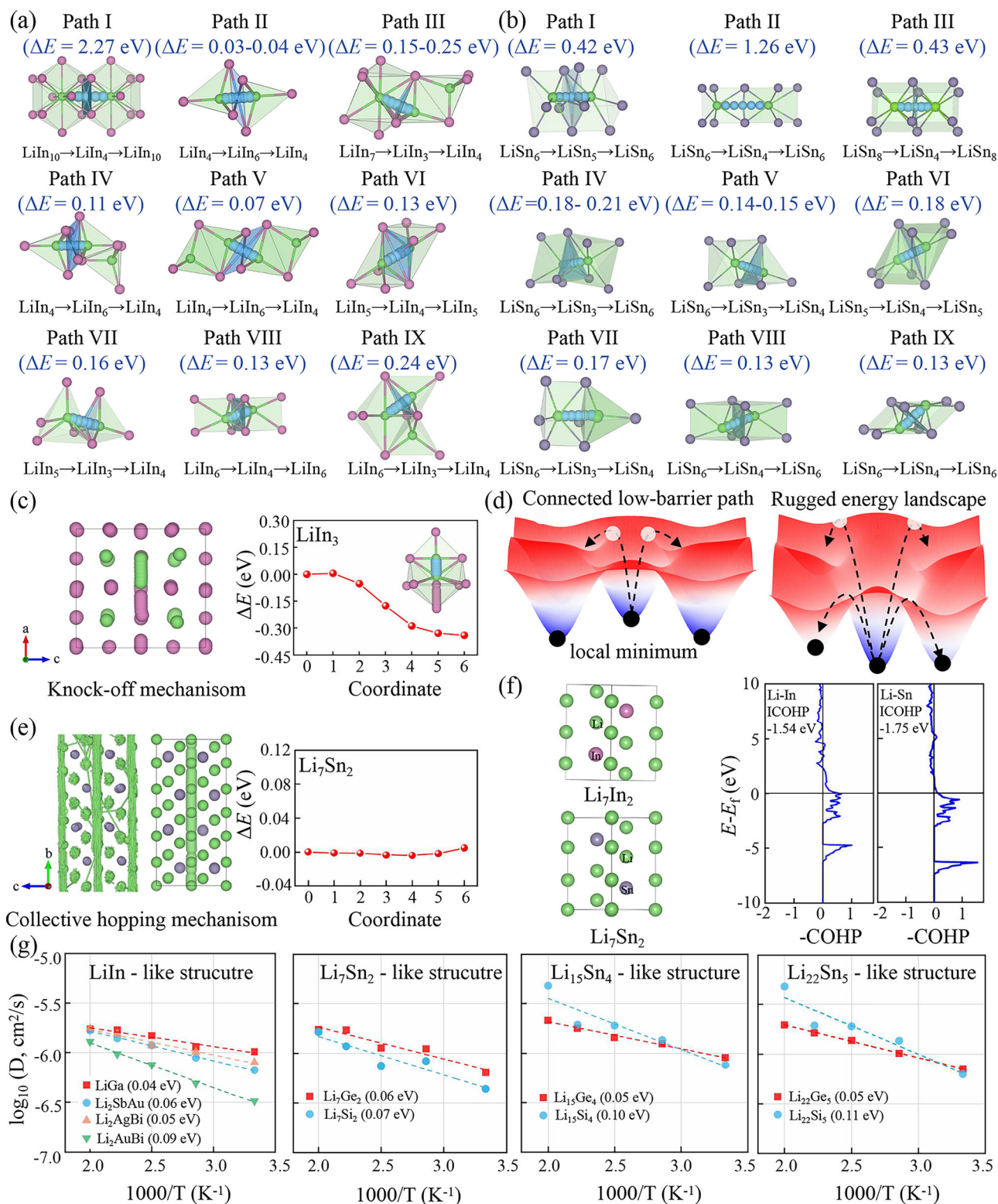


Fig. 4 | Diffusion pathways, energy barriers (ΔE), and bonding interactions in Li-In and Li-Sn alloys. a, b Selected local vacancy diffusion pathways in the stable phases of the Li-In and Li-Sn systems, with Li atoms in green, In atoms in mauve, Sn atoms in purple, and transition-state Li positions in blue. **c** Interstitial diffusion pathways and energy barriers (ΔE) in LiIn_3 . **d** Schematic illustrations of vacancy defect migration pathways in the potential energy landscape for the connected low-barrier pathway and the Rugged energy landscape. The energy difference between a local minimum and nearby saddle point determines the barrier for Li-ion migration.

e Interstitial diffusion pathways and energy barriers in Li_7Sn_2 . **f** Crystal Orbital Hamilton Population (COHP) diagrams of the interactions between In/Sn atoms and their nearest-neighbor Li atoms, where E_f is the fermi energy. The integrated COHP (ICOHP), which reflects the bond strength (the more negative the value, the stronger the bond), is also calculated for Li_7In_2 and Li_7Sn_2 . **g** Temperature dependence of the Li-ion diffusion coefficient in the candidate structures, used to evaluate the generality of favorable diffusion frameworks.

As Li content increases, a continuous Li network enables vacancy diffusion, while structural diversity introduces path-dependent barrier variations (0.03–0.25 eV). Path II (isolated LiIn_4 tetrahedron $\rightarrow$ LiIn_6 octahedron $\rightarrow$ LiIn_4 tetrahedron) exhibits the lowest barrier (0.04 eV in LiIn and 0.03 eV in Li_5In_4) and 3-D and 2-D connectivity, enabling long-range migration. Higher-barrier Path III (LiIn_7 coplanar polyhedron $\rightarrow$ LiIn_3 triangle $\rightarrow$ LiIn_4 tetrahedron), such as 0.24 eV in Li_5In_4 and 0.25 eV in Li_3In_2 , provide connectivity, but restricting diffusion due to high barrier. In Li_5In_4 , Li migration occurs primarily along Path II with 2-D connectivity. In Li_3In_2 , diffusion relies on Path III, reducing the diffusion coefficient from $6.3 \times 10^{-7} \text{ cm}^2/\text{s}$ in Li_5In_4 to $1.6 \times 10^{-8} \text{ cm}^2/\text{s}$. This trend aligns with experimental voltage curves: around $x \approx 1.25$, the lithiation profile of Li-In alloys shifts from a smooth plateau to pronounced fluctuations, reflecting decreased interface stability and a transition toward Li metal deposition⁵⁹. In Li_3In , only Path IX (shared edge $\text{LiIn}_4 \rightarrow \text{LiIn}_6$, $\Delta E = 0.24$ eV) remains, further reducing the diffusion coefficient to $2.9 \times 10^{-8} \text{ cm}^2/\text{s}$. Other low-barrier paths, such as Path V (0.07 eV), exist in isolation and contribute only to local hopping, with negligible effect on overall diffusion (Fig. S18).

Taken together, these results demonstrate that Li-In diffusion is determined not only by the energy barriers of individual paths but, more critically, by the connectivity of low-energy paths. Figure 4d illustrates vacancy-mediated hopping in the potential energy landscape. When low-barrier pathways are well-connected, Li atoms can diffuse along these continuous paths, markedly enhancing long-range diffusion. In contrast, when low-barrier paths are isolated or the potential energy landscape is rugged, Li ions become trapped in local minima, significantly limiting overall mobility.

Compared to Li-In, Li-Sn exhibits both similarities and distinct differences in diffusion mechanisms (Figs. S19–S20). At low Li content, interstitial Li dominates in both systems; as Li content increases, diffusion gradually shifts to vacancy mediation. However, in high-Li phases of Li-Sn (e.g., Li_7Sn_2), diffusion reverts to interstitial dominance, a feature not observed in Li-In compounds. Calculations show that in Li_7Sn_2 , the diffusion coefficient reaches $8.4 \times 10^{-7} \text{ cm}^2/\text{s}$ —the highest among stable Li-Sn phases. NEB and trajectory analysis (Fig. 4e) reveal that this is due to dense Li chains: the introduction of interstitial Li induces cooperative migration of multiple Li atoms along the chain through Coulombic interactions, forming a low-energy collective hopping mechanism that significantly enhances ionic mobility.

Under different carrier conditions, metastable phases of Li-In and Li-Sn further corroborate these mechanisms (Figs. S20–S21). Specifically, $\text{Li}_{13}\text{In}_5$ (*Pm*) is vacancy-mediated, Li_7In_2 , $\text{Li}_{22}\text{Sn}_5$, and $\text{Li}_{15}\text{Sn}_4$ are interstitial-mediated, and all exhibit diffusion coefficients on the order of $10^{-7} \text{ cm}^2/\text{s}$, demonstrating high ionic conductivity. No vacancy-mediated phases in the Li-Sn system achieve such high diffusion coefficients. From the perspective of local diffusion configurations, no ultra-low-barrier pathway analogous to Path II in LiIn and Li_5In_4 is observed in Li-Sn (Fig. 4b). In contrast, Path IV and Path VIII in Li-Sn closely resemble the corresponding paths in the Li-In system and exhibit comparable migration barriers, indicating that these diffusion modes are only weakly dependent on the chemical identity of the M atom. For structurally analogous pairs (Li_7Sn_2 vs. Li_7In_2 , Li_5Sn_2 vs. Li_5In_2), Li-In systems consistently show slightly higher diffusion coefficients, reflecting weaker Li-In interactions compared to Li-Sn, as revealed by the crystal orbital Hamilton population (COHP) analysis⁶¹ (Fig. 4f). Nevertheless, the overall Li diffusion behavior (Fig. S22) remains highly similar across the two systems, highlighting the critical role of crystal structure in dictating diffusion properties. In addition, *Cmmm*- Li_7Sn_2 exhibits a collective hopping mechanism, with a calculated migration barrier lower than that of the previously reported knock-off mechanism (Fig. S23)⁶⁰, thereby identifying a lower-barrier diffusion pathway that may be operative in other Li-rich alloys. This result further confirms the capability of NEP-MD simulations to explore a wide range of Li diffusion pathways, enabling the discovery of previously unrecognized diffusion behaviors.

To assess the generality of favorable diffusion frameworks, we selected Li-rich intermetallics with similar Li content from Materials Project and

OQMD: LiIn-type (LiGa , Li_2SbAu , Li_2AgBi , Li_2AuBi), Li_7Sn_2 -type (Li_7Ge_2 , Li_7Si_2), $\text{Li}_{15}\text{Sn}_4$ -type ($\text{Li}_{15}\text{Ge}_4$, $\text{Li}_{15}\text{Si}_4$), and $\text{Li}_{22}\text{Sn}_5$ -type ($\text{Li}_{22}\text{Ge}_5$, $\text{Li}_{22}\text{Si}_5$) (Fig. S24). NEP potentials were first validated with DFT (Fig. S25), and machine-learning accelerated MD simulations showed diffusion coefficients around $10^{-7} \text{ cm}^2/\text{s}$, consistent with Li-In and Li-Sn structures. This demonstrates strong structural inheritance and generalizability: connectivity and local coordination primarily determine Li mobility, while chemical composition fine-tunes performance.

Building on this mechanistic insight, we used a Gradient Boosting Regressor (GBR) model to quantify factors controlling local Li diffusion, incorporating descriptors, such as migration distance (d), charge redistribution between initial and transition states ($|e|$), Li-In bond length (L_{ave}), coordination number (N), polyhedral volume (V), and energy change ($\Delta E_{\text{IS-ES}}$), defined as the difference in energy between the initial and final states along the Li migration pathway. Minimal descriptor correlation (Pearson < 0.75 , Fig. 5a) ensures distinct contributions. GBR predictions closely match DFT barriers (validation/test RMSE = 0.04/0.06 eV, $R^2 = 0.77/0.85$, Fig. 5b), outperforming random forest regression (RFR)⁶² (Fig. 5c). Feature importance analysis identifies migration distance and charge redistribution as dominant factors, and the scatter reflects that the overall trend: longer distances and larger charge redistribution correlate with higher barriers (Fig. 5d, S26). For example, in LiIn_3 , PATH I exhibits a large charge reorganization ($|e| = 0.102 e^-$), leading to a high migration barrier (2.27 eV), whereas PATH II in Li_5In_4 shows smaller charge redistribution ($|e| = 0.002 e^-$), resulting in a low barrier of only 0.03 eV (Fig. S27). Similarly, for the same local configuration but different compositions (e.g., PATH III in Li_5In_4 and Li_3In_2), the degree of charge reorganization remains comparable, indicating that the local atomic environment, rather than the overall stoichiometry, primarily governs the charge redistribution and thus the migration barrier. Furthermore, variations in tetrahedral connectivity and jump distances lead to different barriers even for structurally similar LiIn_4 – LiIn_6 – LiIn_4 diffusion processes (Fig. 4e). Identical local diffusion pathways across compositions exhibit comparable structural and electronic characteristics, resulting in similar migration barriers (Fig. 5f), confirming the generality of high-diffusion configuration and enabling extrapolation to other Li-rich intermetallics.

Discussion

Our analysis demonstrates that Li-ion transport in Li-In and Li-Sn alloys is governed by three interrelated factors: the Li content, the local atomic environment, and the connectivity of low-energy migration pathways. The connectivity of the Li sublattice network determines the dominant diffusion mechanism: isolated Li sites promote interstitial-mediated diffusion, whereas continuous Li networks favor vacancy-mediated diffusion. Local atomic environments, including Li-Li distances and charge redistribution along hopping paths, modulate migration barriers for individual steps. For instance, in Li_5In_4 , diffusion along well-connected 2-D pathways results in high Li mobility ($D \approx 6.3 \times 10^{-7} \text{ cm}^2/\text{s}$), whereas in Li_3In_2 , reliance on higher-barrier, less-connected paths reduces the diffusion coefficient by more than an order of magnitude. These results clarify that effective long-range Li mobility arises from the interplay between local barriers and the connectivity of accessible low-energy pathways.

Chemical composition (Li-In vs. Li-Sn) fine-tunes local migration barriers without fundamentally altering the dominant diffusion mechanism, which is determined by Li network connectivity. COHP analyses indicate that Li-Sn interactions are slightly stronger than Li-In, leading to marginally higher local barriers, consistent with the modest differences in diffusion coefficients between analogous phases. In high-Li phases of Li-Sn (e.g., Li_7Sn_2), dense Li chains induce cooperative multi-ion hopping, enabling interstitial-dominated diffusion even at high Li content and maximizing ionic mobility.

To test the generality of these insights, we examined chemically related Li-rich intermetallics with comparable Li content, including LiIn-type (LiGa , Li_2SbAu , Li_2AgBi , Li_2AuBi), Li_7Sn_2 -type (Li_7Ge_2 , Li_7Si_2), $\text{Li}_{15}\text{Sn}_4$ -type ($\text{Li}_{15}\text{Ge}_4$, $\text{Li}_{15}\text{Si}_4$), and $\text{Li}_{22}\text{Sn}_5$ -type ($\text{Li}_{22}\text{Ge}_5$, $\text{Li}_{22}\text{Si}_5$) frameworks.

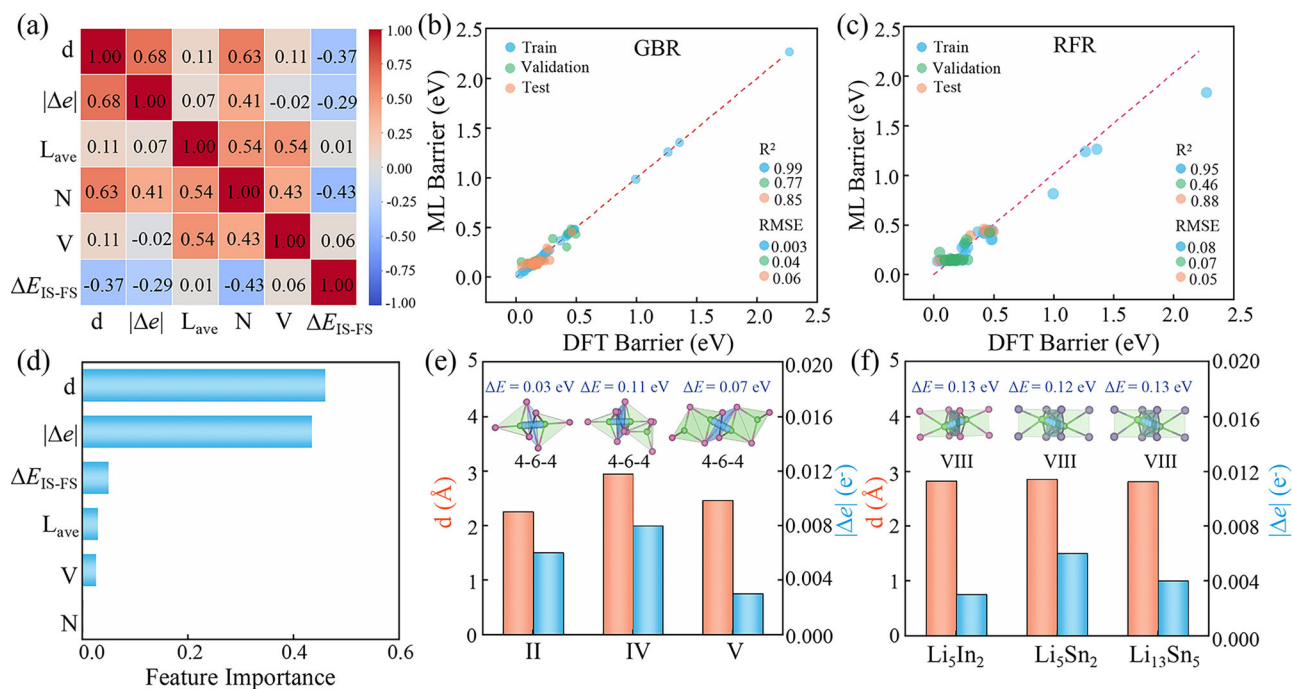


Fig. 5 | Machine-learning (ML) prediction and feature analysis of Li-ion migration in Li-In and Li-Sn systems. **a** Heat map of the Pearson correlation coefficient matrix among the selected structural and chemical descriptors used for machine-learning modeling, including migration distance (d), charge redistribution ($|\Delta e|$), average Li-In bond length (L_{ave}), coordination number (N), polyhedral volume (V), and energy change (ΔE_{IS-FS}). **b, c** Comparison of Li migration energy barriers obtained from DFT with those predicted by Gradient Boosting Regressor

(GBR) model and random forest regression (RFR) for both training and testing datasets of Li-In and Li-Sn system. **d** Feature importance analysis from the GBR model for predicting migration barriers. **e** Li migration distances and local charge variations in configurations II, IV, and V in Li_5In_2 and Li_2In . **f** Li migration distances and local charge variations in configurations VIII in Li_5In_2 , Li_5Sn_2 and $Li_{13}Sn_5$. Li atoms are shown in green, In atoms in mauve, Sn atoms in purple, and transition-state Li positions in blue.

NEP-driven MD simulations predict diffusion coefficients on the order of 10^{-7} cm²/s, consistent with Li-In and Li-Sn systems. This similarity is not coincidental. These compounds share analogous local environments and Li sublattice connectivity that facilitate low-energy hopping, confirming that the key factors controlling Li mobility, including carrier type, local coordination, and connectivity of migration pathways, are transferable across chemistries.

Overall, our study establishes three central principles governing Li transport in complex alloy anodes: (i) the connectivity of the Li sublattice network determines the dominant carrier type, (ii) the local atomic environment sets migration barriers for individual hopping steps, and (iii) continuous networks of favorable migration paths enable high long-range mobility. By identifying structural configurations that consistently support fast Li diffusion, this framework provides a mechanistic basis for screening Li-rich intermetallics, facilitating the rational design of next-generation alloy electrodes with enhanced ionic conductivity.

Methods

DFT Calculation

In this study, DFT calculations are performed using the Vienna Ab initio Simulation Package (VASP) to obtain energies, atomic forces, and virial tensors required for training NEPs⁶³. The projector augmented-wave (PAW) method is employed in conjunction with the Perdew-Burke-Ernzerhof (PBE) exchange-correlation functional within the generalized gradient approximation (GGA)⁶⁴. A plane-wave energy cutoff of 500 eV is used. The Brillouin zone is sampled using a Monkhorst-Pack k-point grid with a spacing of $0.168 \text{ \AA}^{-1}$. The energy convergence criterion for static calculations is set to 10^{-5} eV, and the force convergence threshold is $0.01 \text{ eV/\AA}$ to ensure high accuracy in energy, forces, and structural details. Phonon calculations are carried out using the finite displacement method⁶⁵, implemented in the PHONOPY code⁶⁶. COHP⁶¹ calculations are performed with the LOBSTER software package. The PHONOPY

package is also used to calculate ZPE corrections. The climbing-image nudged elastic band (CI-NEB) method is employed to investigate the influence of different local chemical environments on Li diffusion pathways and energy barriers¹⁹.

For the resulting stable structures, the average voltage is computed from the total DFT energies using the following formula:

$$V(x_1, x_2) = - \frac{E(Li_{x_1}M) - E(Li_{x_2}M) - (x_1 - x_2)E(Li)}{(x_1 - x_2)} \quad (1)$$

AIMD Calculations

AIMD simulations in the canonical ensemble (NVT) are performed using a Nosé-Hoover thermostat to control temperature, ranging from 300 to 1800 K and are used to generate the initial training dataset. The minimal supercell approach is adopted to ensure that the lattice vectors exceeded $8 \text{ \AA}$ in all directions. A time step of 1 fs is used, with simulation durations of at least 5 ps. For AIMD simulations, only the Γ -point is used for Brillouin zone sampling.

NEP-Based Structure Search

During the CALYPSO-NEP structural exploration to generate dataset, we performed variable-stoichiometry searches within Li_xM_y ($M = \text{In or Sn}$; $x = 1-20$; $y = 1-20$) and the total number of atoms capped at 40 per unit cell. In each iteration, 2500 low-energy structures were generated and then reduced by farthest-point sampling to retain only representative ones. This iterative process was repeated until the NEP model achieved full consistency with DFT energies and forces.

After obtaining a high-accuracy NEP, both CALYPSO⁴⁶ and USPEX⁶⁷ were employed to explore stable and metastable phases in the Li-In and Li-Sn systems. For fixed-stoichiometry searches in CALYPSO, the simulation cell size was set to 1-4 formula units. After the first generation, 50% of the

low-energy structures were selected to construct the next generation via the particle-swarm optimization (PSO) algorithm. In most cases, the structural search was terminated after approximately 4000–5000 structures had been generated.

The USPEX code combined with the NEP is used to predict stable phases in the Li–In and Li–Sn alloy systems. A variable-composition search is first performed, where the first generation consisted of 60 structures, and the number of atoms per primitive cell is limited to between 2 and 36. For the generation of subsequent structures (each generation containing 200 structures), 40% of the lowest-energy structures from the previous generation are optimized using variation operators-including heredity, softmutation, transmutation, and lattice mutation-while the remaining 60% are randomly generated. This process is repeated for 101 generations.

Based on the convex hull results from the variable-composition search, a fixed-composition search is then conducted on more than 60 low-energy Li–In/Sn stoichiometries. Each generation consisted of 60 structures, and the search is repeated for 50 generations. In each generation, 50% of the structures are randomly generated and the remaining 50% are produced through variation operators applied to existing low-energy configurations.

Large-Scale Molecular Dynamic (MD) Simulations

To ensure that the NEP potential accurately describes Li diffusion in the predicted phases, MD simulations are performed, with a heating phase (from 0 to 650 K, lasting 1 ns) followed by equilibration phase (lasting 1 ns). During the simulation, structures are continuously collected, and configurations with the largest energy and force deviations are added to the training dataset.

The diffusion coefficient calculation are carried out using the NEP-based Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) software package⁶⁸. The lattice constants of the structures used for ion diffusion coefficient calculations is exceeding 30 Å. The defective structures are generated using the Atomsk tool by randomly adding or removing approximately 1% of the atoms in the atomic structure⁶⁹. The simulations are conducted in the temperature range from 300 to 500 K, with a 50 K interval. For each temperature, the system is heated for 1 ns and equilibrated for 5 ns under the NPT ensemble with a time step of 1 fs. The simulation results are visualized and analyzed using the Open Visualization Tool (OVITO) and Visual Molecular Dynamics (VMD) software^{70,71}. The diffusion coefficient of Li atom is calculated through the MSD over time intervals.

$$D = \frac{1}{2Nd\Delta t} \sum_{i=1}^N \langle |r_i(r + \Delta t) - r_i(t)|^2 \rangle_t \quad (2)$$

where N is the total number of diffusing ions, $d = 3$ represents the dimension of the diffusion system, $r_i(t)$ is the displacement of the i -th ion at time t , and $\langle \dots \rangle$ denotes the time average. The initial ballistic regime is excluded from our analysis. Only the time interval, where the MSD exhibits a clear linear dependence on time, is selected for extracting diffusion coefficients.

The activation energy (E_a) for ion diffusion is calculated using the Arrhenius equation.

$$\sigma = \sigma_0 \times \exp\left(\frac{-E_a}{k_B T}\right) \quad (3)$$

where σ_0 is the pre-exponential factor of the conductivity, k_B is the Boltzmann constant, and T is the temperature.

Gradient Boosting Regression (GBR)

In this study, GBR model is employed to analyze the key factors influencing local Li diffusion. GBR is more suitable for a small database and exhibits good performance in many applications^{72–74}. The model is implemented using the Python package scikit-learn with tuned hyperparameters. A DFT calculated dataset of 90 points from Li–In and Li–Sn compositions is used to generate the training, testing and validation data by a random split of data.

The feature importance and predictive accuracy (RMSE and R^2) are evaluated.

Data availability

The datasets generated and analyzed during the current study are presented as figures and tables in the manuscript and Supplementary information and are available from the corresponding authors on reasonable request.

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Author contributions

D.J. performed calculations, analyzed the data, and wrote the original draft. S.D., H.Q., Y.L., and X.L. assisted the calculations, and participated in the data analysis. G.Y. conceived the initial idea, edited and reviewed the

manuscript, and supervised the project. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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